

ATIA: A Link between Inflammation and Hypoxia

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In this issue of *Molecular Cell*, Choksi et al. (2011) report the identification of an NF- κ B-independent ATIA (anti-TNF α -induced apoptosis)-Thioredoxin 2 axis that inhibits TNF α - and hypoxia-induced apoptosis through elimination of excessive reactive oxygen species directly.

The current paradigm for inhibition of TNF α -induced apoptosis centers on NF- κ B, which inhibits caspases and prevents prolonged JNK activation. The antiapoptotic effect of NF- κ B has also been suggested to associate with the elimination of excessive reactive oxygen species (ROS). Choksi et al. (2011) have now shown that ATIA (anti-TNF α -induced apoptosis) inhibits TNF α -induced apoptosis independently of NF- κ B through TRX2-mediated elimination of excessive ROS. The findings provide a novel paradigm for inhibition of apoptosis induced by TNF α and other death signals by controlling ROS accumulation directly.

The pleiotropic inflammatory cytokine TNF α regulates immune responses, inflammation, proliferation, and cell death (apoptosis and necrosis), and its regulation of apoptosis is mainly mediated by its membrane receptor 1 (TNF-R1). Upon TNF α stimulation, TNF-R1 trimer recruits multiple adaptors such as TRADD, TRAF2, TRAF5, RIP1, cIAP1 and cIAP2 and other modulators or regulators such as Miz1 and the linear ubiquitin chain assembly complex (Gerlach et al., 2011, Liu and Lin, 2007; Liu et al., 2009), resulting in activation of multiple downstream effectors including NF- κ B, JNK, and caspases. TNF α -induced apoptosis is tightly regulated by the interplay between NF- κ B and the other downstream effectors (Liu and Lin, 2007). NF- κ B induces expression of various inhibitors of apoptosis (IAPs) including cFLIP_L, thereby inhibiting caspases. NF- κ B also prevents prolonged JNK activation (De Smaele et al., 2001; Tang et al., 2001). When NF- κ B activation is impaired, prolonged JNK activation activates the E3 ligase Itch, which ubiquitinates cFLIP_L and promotes its proteasomal degradation. This

releases cFLIP_L-mediated inhibition on caspase 8, which is involved in the initiation of TNF α -induced apoptosis (Chang et al., 2006). Several mechanisms have been proposed to explain how NF- κ B prevents prolonged JNK activation, including the induction of antioxidants that eliminate excessive ROS, which is a strong JNK activator. While inhibition of TNF α -induced apoptosis by NF- κ B-dependent mechanism has been intensively studied, how TNF α -induced apoptosis is regulated by NF- κ B-independent mechanism is less clear.

To begin to address this issue, Choksi et al. (2011) used TRAF2-deficient murine embryonic fibroblasts (MEFs), which are sensitive to TNF α cytotoxicity despite having NF- κ B activation, as a model system to search for gene(s) that protect the cells from TNF α -induced apoptosis. The authors convincingly show that ATIA, also known as vasin, inhibits TNF α -induced apoptosis, but not necrosis through elimination of ROS. Upon TNF α stimulation, ROS is generated on both cell membrane, which is associated with necrosis, and mitochondria, which is associated with apoptosis (Kim et al., 2007). The authors provide strong evidence that the link between ATIA and ROS is the ATIA-interacting protein TRX2, a mitochondrial-specific thioredoxin protein that is involved in cell survival through elimination of excessive ROS generated on mitochondria (Tanaka et al., 2002). The authors elegantly demonstrate that ATIA is required for maintaining TRX2 at its reduced form, which is active in eliminating excessive ROS and thereby inhibiting apoptosis. As expected, ATIA expression is dependent on TRAF2, a known regulator of ROS (Lin et al., 2004). However, ATIA

expression is not induced by TNF α . The authors suggest that TRAF2-dependent basal level expression of ATIA may be sufficient in maintaining TRX2 at its reduced form. The authors provide strong evidence that the antiapoptotic effect of ATIA is independent of NF- κ B. The loss of ATIA does not affect NF- κ B activation or the kinetics of JNK activation. Conversely, RelA null MEFs have high-level expression of ATIA, although the cells are highly sensitive to TNF α -induced apoptosis. These findings clearly demonstrate that both NF- κ B and ATIA are required for cell survival in response to TNF α but independently from each other. Since excessive ROS contributes to apoptosis induced by many death stimuli, it is not surprising that ATIA also inhibits apoptosis induced by other death stimuli. The list includes hypoxia, ultraviolet, and H₂O₂. The authors clearly show that unlike TNF α , hypoxia significantly upregulates ATIA expression via HIF-1 α , and demonstrate that ATIA is a HIF-1 target gene. According to the model proposed by the authors (Figure 1), ATIA functions as a convergent point for various extracellular stimuli including TNF α and hypoxia to inhibit apoptosis and ATIA itself is regulated by different mechanisms. The inhibition of both TNF α - and hypoxia-induced apoptosis by ATIA could have important pathophysiological implications. The authors provide strong evidence that genetic disruption of ATIA in mouse results in massive spermatocyte apoptosis and male infertility. Furthermore, ATIA expression is significantly upregulated in human glioblastoma and is required for the tumor cells survival. It is known that solid tumors need to survive and grow under hypoxia conditions and in some cases to withstand

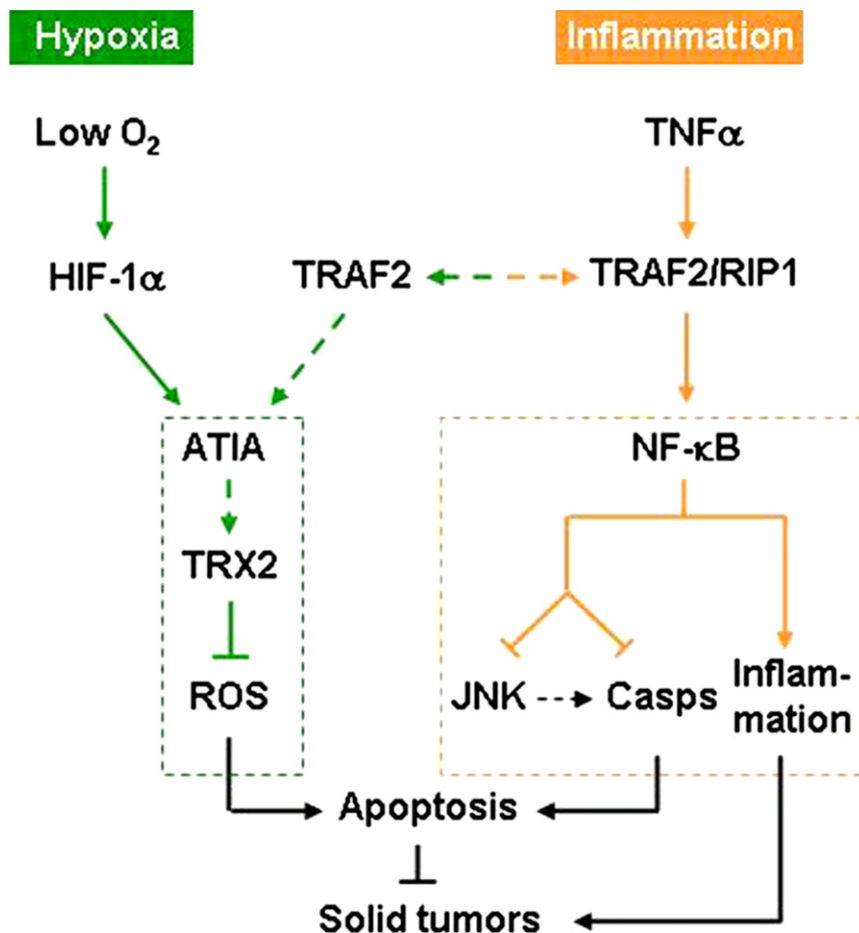


Figure 1. Schematic Illustration of ATIA Serving as a Link between Inflammation and Hypoxia

The ATIA-TRX2 axis inhibits apoptosis induced by both $\text{TNF}\alpha$ and low O_2 through elimination of excessive ROS on mitochondria. Thus, ATIA may be a key regulator in solid tumors like human glioblastoma, which needs to survive and grow under hypoxia conditions and withstands $\text{TNF}\alpha$ cytotoxicity.

$\text{TNF}\alpha$ cytotoxicity associated with acute inflammation. By inhibiting both $\text{TNF}\alpha$ - and hypoxia-induced apoptosis, ATIA may not only protect glioblastoma from death insults, but may also allow the tumor cells to benefit from the tumor promoting effect of low-dose $\text{TNF}\alpha$ that is often associated with chronic inflammation. Thus, ATIA may act as a key regulator in certain type of solid tumors such as human glioblastoma and could be used as a potential diagnostic biomarker and a therapeutic target.

The findings by Choksi et al. (2011) shed light on NF- κ B-independent mecha-

nism that plays a critical role in inhibition of apoptosis induced by $\text{TNF}\alpha$, hypoxia, and other death signals. More importantly, it uncovers a link between inflammation and hypoxia. However, several interesting questions remain to be answered. ATIA localizes on both cell membrane (mem-ATIA) and mitochondria (mito-ATIA), while only the latter inhibits $\text{TNF}\alpha$ -induced apoptosis. Does ATIA redistribute between cell membrane and mitochondria upon stimulation by $\text{TNF}\alpha$, hypoxia, and other death signals? How does TRAF2 regulate basal level expression of ATIA? How does ATIA regulate

the redox status of TRX2? Given the fact that ATIA controls ROS generation independently of NF- κ B and that the loss of ATIA has no detectable effects on the kinetics of JNK activation, the role of ROS in $\text{TNF}\alpha$ -induced prolonged JNK activation may need to be further investigated. Why is only spermatocyte apoptosis observed in ATIA knockout mice? Is ATIA upregulated in other kinds of human tumors in addition to glioblastoma? Since inflammation and hypoxia are closely associated with tumor initiation and progression, investigation of the molecular mechanism underlying the regulation of ATIA will not only shed light on the link between inflammation and hypoxia, but may also provide novel rationale and potential therapeutic targets for prevention and treatment of certain type of solid tumors like human glioblastoma.

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